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In [1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
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In [2]: G = 6.674*10**-11
M = 5.974*10**24
m = 7.348*10**22
R = 3.844*10**8
w = 2.662*10**-6
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In [3]: def f(x):
return(G*M/x**2-G*m/(R-x)**2-w**2*x)
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In [4]: def secant(f, guess, delta=1, tolerance = 2**-32):
x0 = guess
x1 = x0 + delta
n = 0
while abs(f(x1))>tolerance:
x2 = x1 - (x1-x0)/(f(x1)-f(x0))*f(x1)
x0 = x1
x1 = x2
n += 1
return(x1, n)
```

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In [13]: secant(f, 1000000, 100)
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Out[13]: (326045071.6618108, 26)
```

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In [ ]:
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